

CLOUD-BASED IMAGE STORAGE APPLICATION

Naan Mudhalvan – Cloud Computing Project

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Year / Semester	2nd Year / Semester 3
College	Annai Fathima College of Arts and Science, Tirumangalam
Cloud Platform	Amazon Web Services (AWS)
AWS Region	Asia Pacific (Mumbai) – ap-south-1
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1. Problem Statement

Create a cloud-based solution for storing and organizing images using cloud storage. The solution should support uploading, storing, organizing, retrieving images, and configuring appropriate access permissions.

2. Objectives

- Upload images to cloud storage.
- Store images securely in Amazon S3.
- Organize images using S3 folders/prefixes.
- Retrieve and open stored images.
- Configure appropriate access permissions.

3. Technologies Used

Technology	Purpose
Amazon S3	Cloud object storage for images
AWS Management Console	Create and manage the S3 bucket
Mobile Web Browser	Perform the AWS implementation
GitHub	Supporting documentation and proof

4. Cloud Architecture

Mobile / User → Amazon S3 Bucket → Folders → Images

Only Amazon S3 is required for this storage-focused implementation. EC2, VPC, database, load balancer, and NAT Gateway are not required.

5. AWS Implementation

5.1 Create S3 Bucket

A general-purpose S3 bucket was created in Asia Pacific (Mumbai), ap-south-1.

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ACLs were disabled using the recommended bucket-owner-enforced setting, and Block Public Access was kept enabled.

5.2 Organize Images

Images are organized using S3 folders/prefixes such as:

- books/
- students/
- products/

5.3 Upload Images

An image was selected from the mobile device and uploaded to the S3 bucket under the books folder.

5.4 Retrieve Images

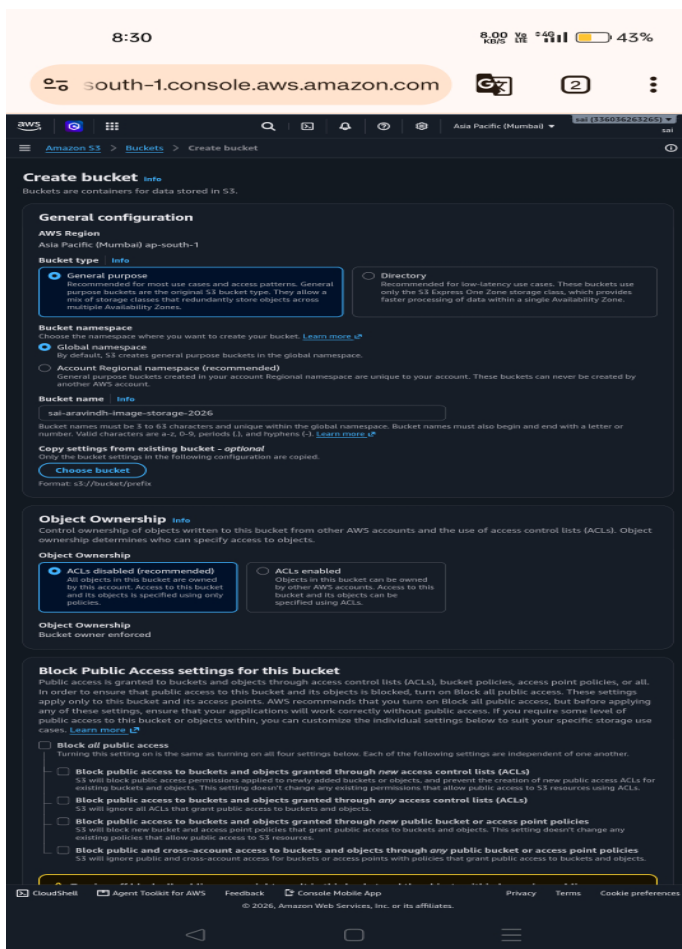
The uploaded image was opened/retrieved from its S3 object location, demonstrating cloud storage and retrieval.

5.5 Access Permissions

Block all public access was kept ON to prevent unintended public access to the bucket and objects.

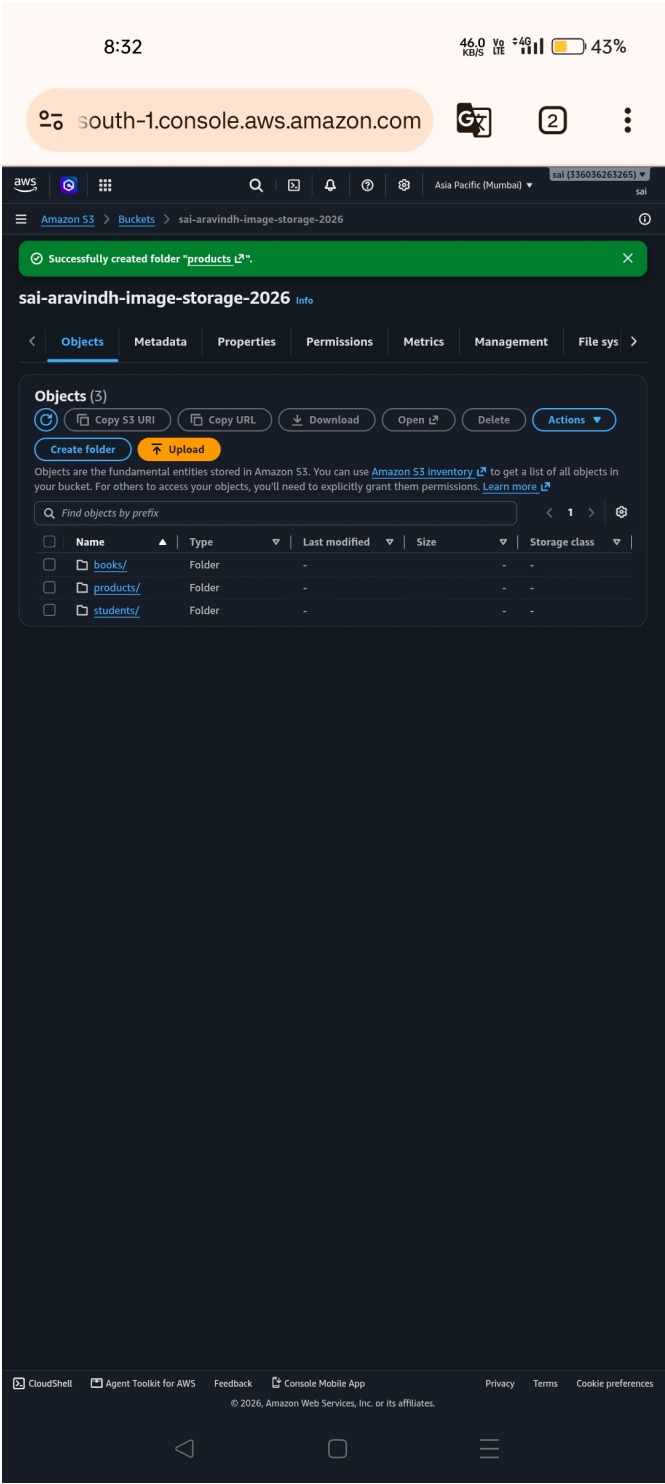
6. Screenshots / Proof

S3 bucket created

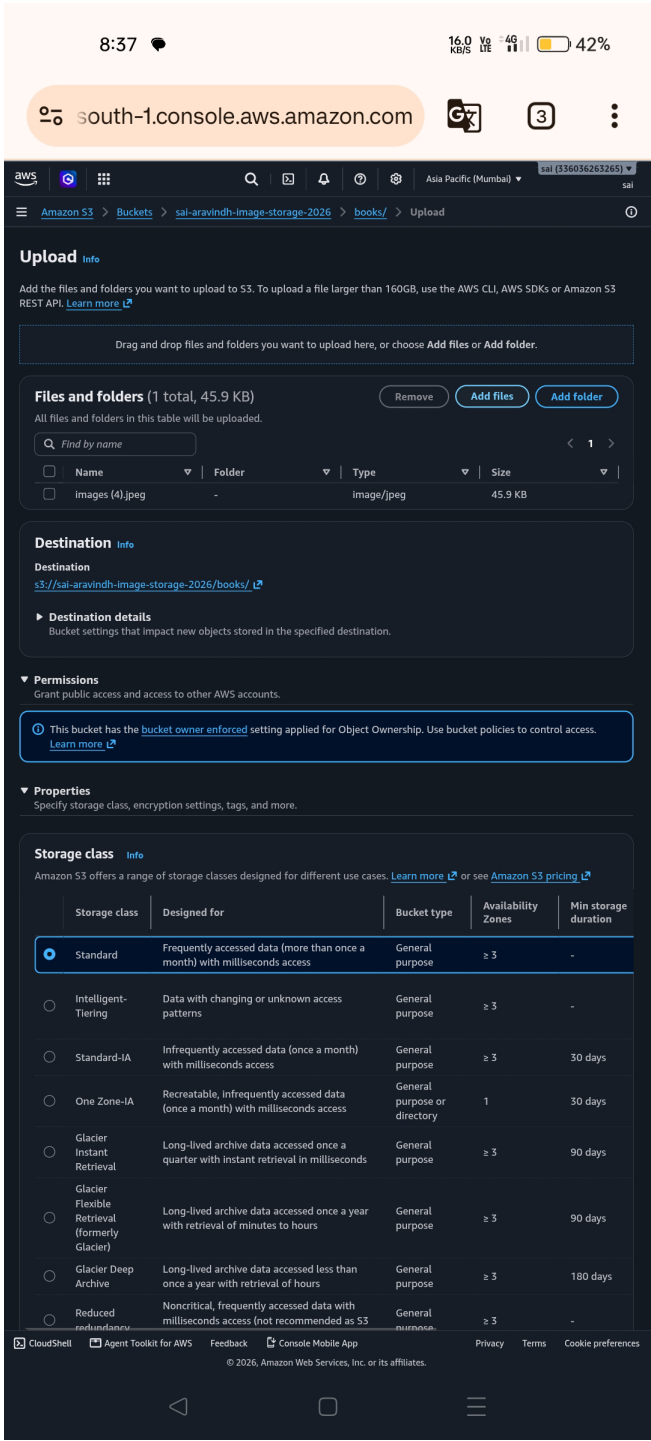


Folder and uploaded image

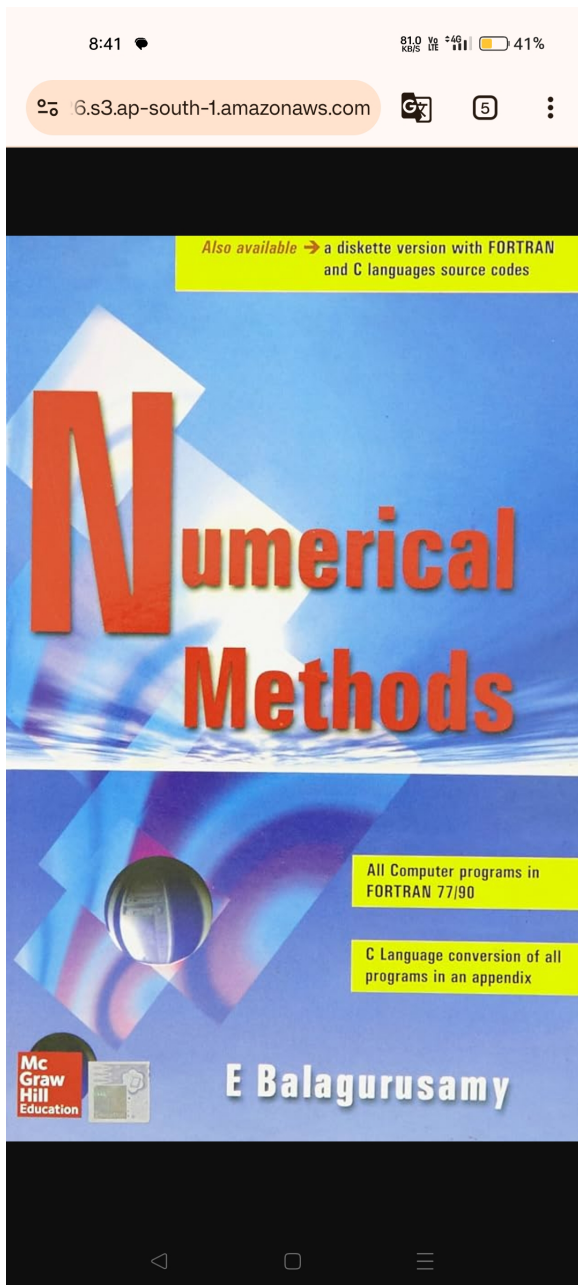
[PASTE AWS SCREENSHOT HERE]



Upload completed / object visible

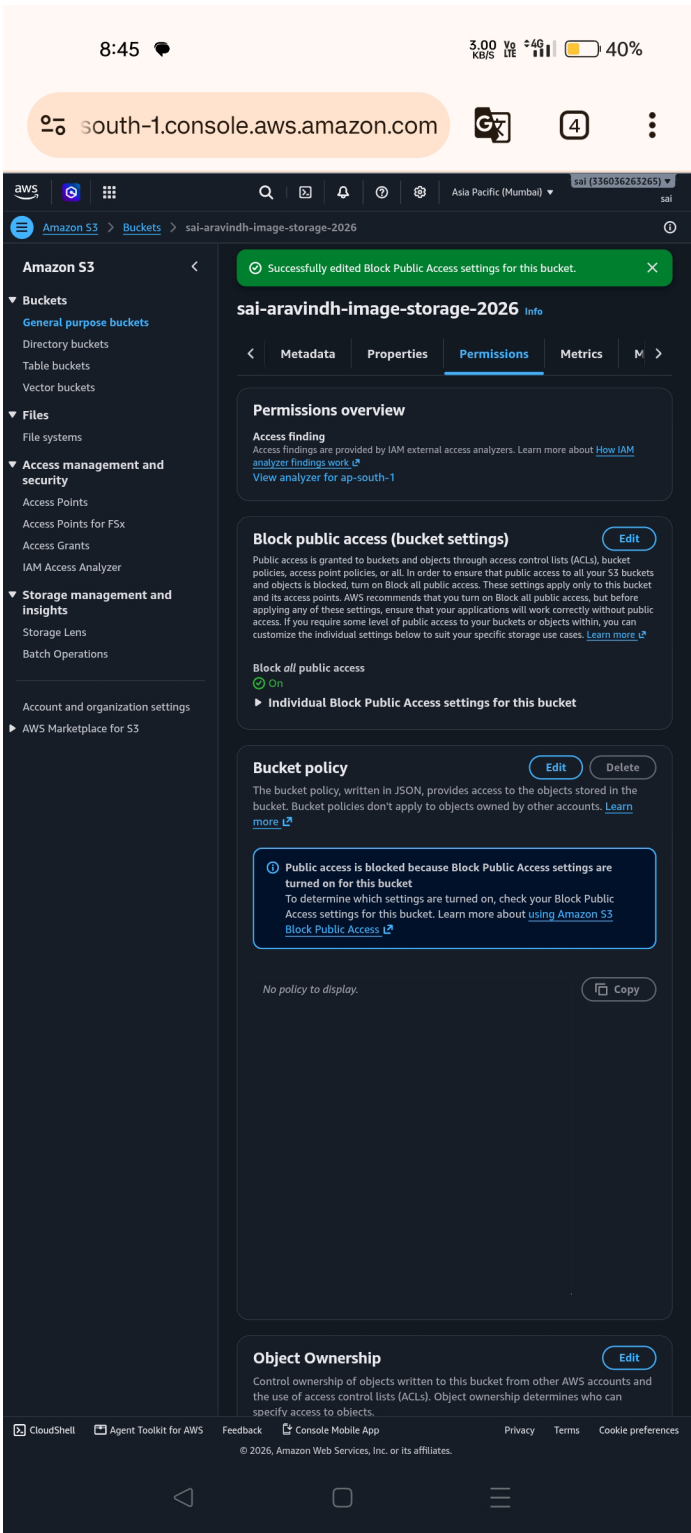


Retrieved or opened image



Permissions showing Block all public access ON

[PASTE AWS SCREENSHOT HERE]



7. GitHub Supporting Proof

GitHub can be used to store the README, screenshots, and project documentation. AWS screenshots remain the primary evidence of the actual cloud implementation.

Suggested repository name: cloud-based-image-storage-aws

8. Advantages

- Simple cloud-based image storage.
- Can be managed from a mobile device.
- Images can be organized using folders/prefixes.
- Access can be controlled using AWS permissions.
- No EC2 server or VPC setup is required for this solution.

9. Future Enhancements

- Create a web/mobile front end.
- Add user authentication.
- Add image search and metadata.
- Add thumbnails and categorization.
- Add backup and lifecycle management.

10. Result

The Cloud-Based Image Storage Application was successfully implemented using Amazon S3 in AWS Mumbai (ap-south-1). Images were uploaded, stored, organized, retrieved, and protected using appropriate access settings.

11. Conclusion

This project demonstrates how Amazon S3 can provide simple and scalable cloud storage for images and satisfy the required upload, storage, organization, retrieval, and access-control features.